

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
 In preparation for General Certificate of Education Advanced Level
 Higher 2

CANDIDATE
NAME

--

CT
GROUP

1	5	S	
---	---	---	--

INDEX
NUMBER

--	--	--	--

BIOLOGY

9648/02

Paper 2 Core Paper

16 September 2016

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
 Write in dark blue or black pen.
 You may use a soft pencil for any diagrams, graph or rough working.
 Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

All working for numerical answers must be shown.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, fasten all your work securely together in 2 separate section A and B.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	80
1	
2	
3	
4	
5	
6	
7	
8	
Section B	20
9 / 10	
Total	

This document consists of **25** printed pages including the cover page and **1** blank page.

[Turn Over]

Section A

Answer **all** questions in this section.

- 1 Fig 1.1 shows some plant cells undergoing mitosis.

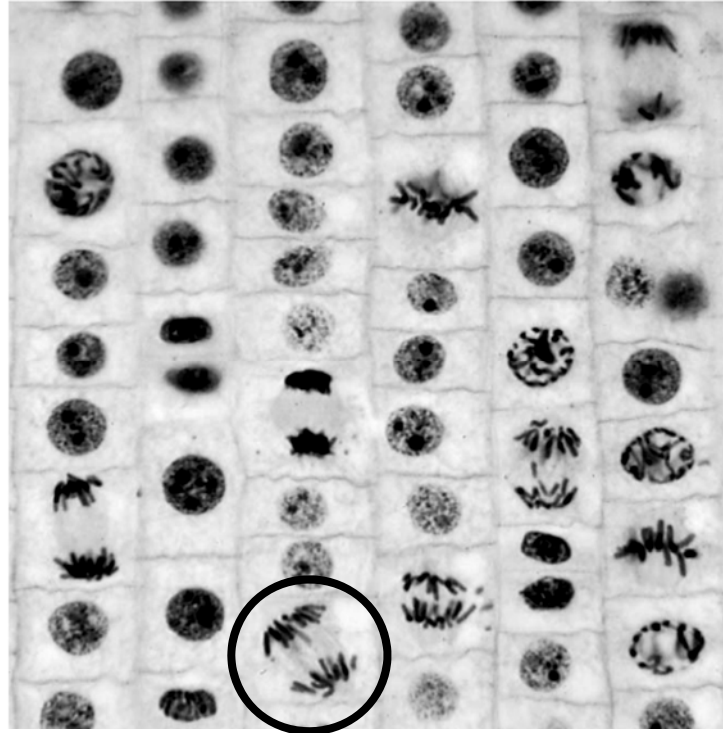


Fig. 1.1

Fig. 1.2 shows the changes in amount of DNA at different stages of the plant life cycle.

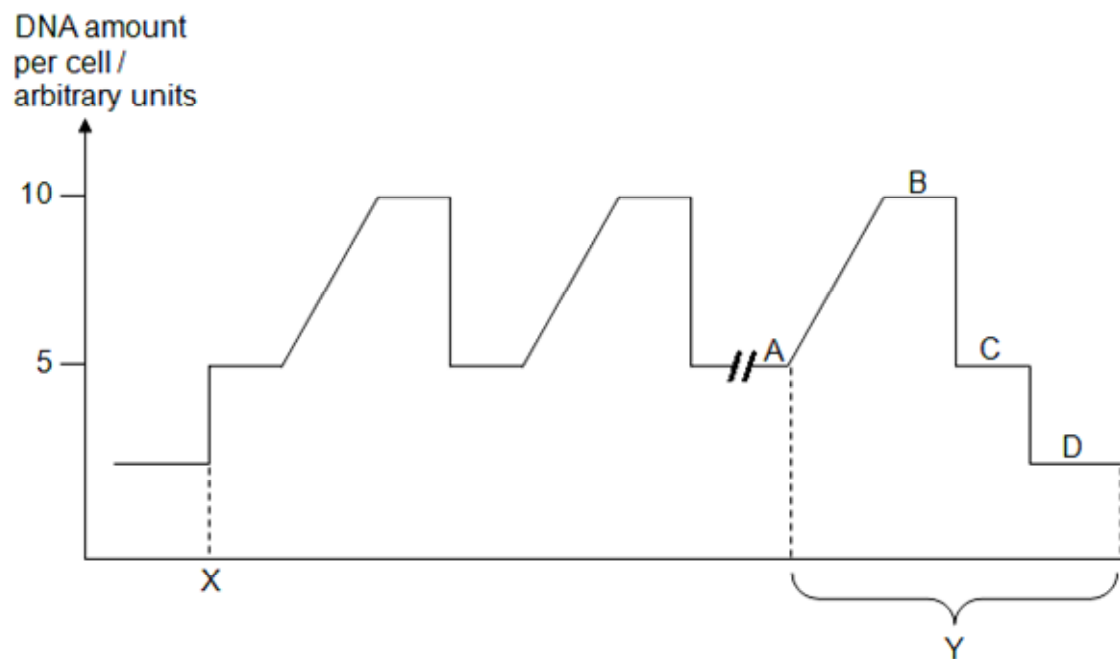


Fig. 1.2

(a) Mark out with an arrow ↓ clearly on Fig. 1.2 which part of the graph corresponds to the stage circled in Fig. 1.1. [1]

(b) From stages **A** to **D** in Fig. 1.2, state all stages

(i) that has/have the same number of chromosomes as shown in Fig. 1.1;

..... [1]

(ii) that has/have the **different** number of chromosomes as shown in Fig. 1.1;

..... [1]

(c) Explain the significance of the stages in **Y** in genetic variation.

.....

 [4]

(d) Explain the significance of the event occurring at **X**.

.....

 [1]

[Total: 8]

[Turn Over]

2 Fig. 2.1 shows a bacterial cell during protein synthesis.

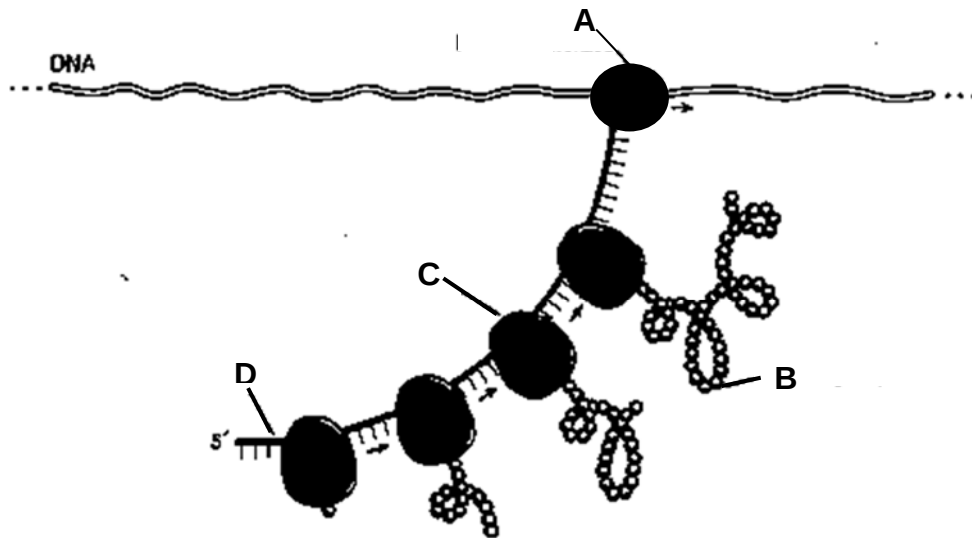


Fig. 2.1

(a) Identify A – D.

A

B

C

D [2]

(b) With reference to Fig. 2.1, suggest evidences that indicate that these processes occurred in a prokaryotic cell.

.....

 [1]

(c) Briefly describe how structure **A** differs from structure **C**.

.....

.....

.....

.....

..... [2]

(d) Activated tRNA can be seen entering structure **C**. Explain how the correct amino acid is joined to a tRNA.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 9]

[Turn Over

- 3 (a) Fig. 3.1 shows the viral load in the blood and CD4+ T cell counts over the course of a typical HIV infection.

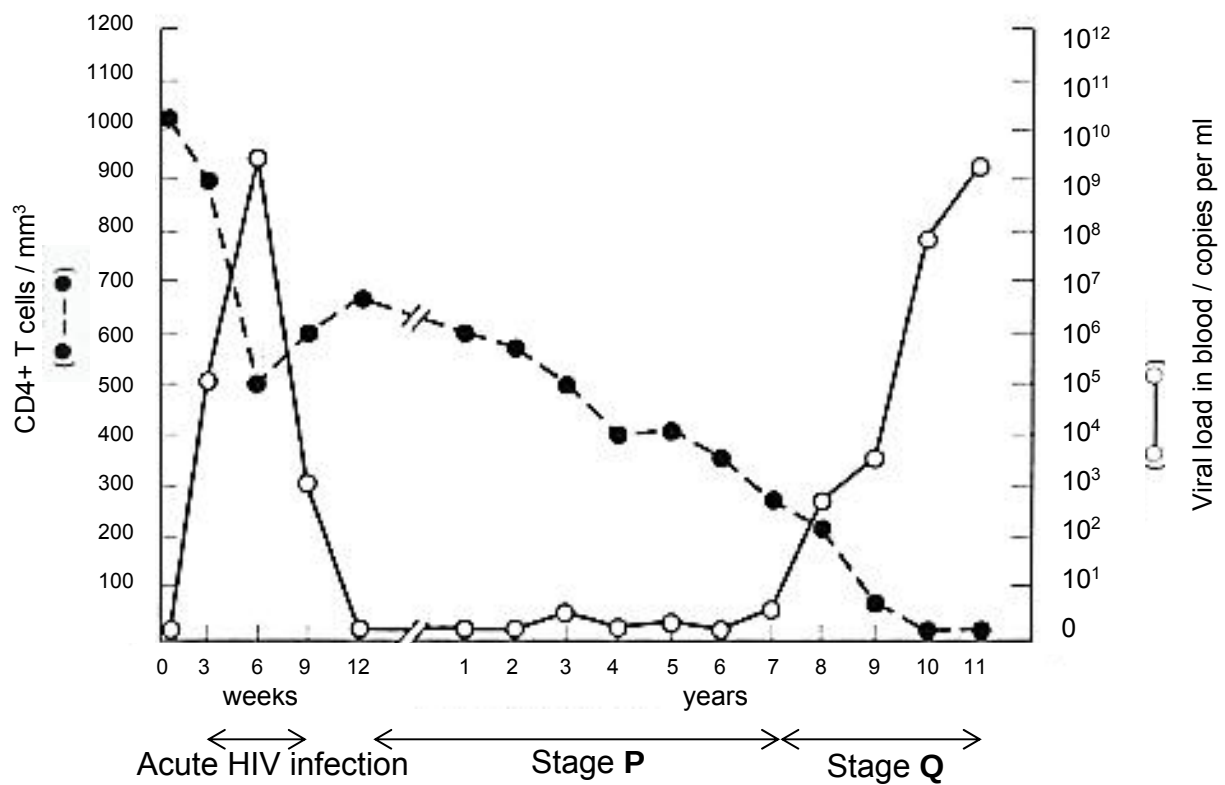


Fig. 3.1

- (i) Many cells in Stage P were found to contain viral DNA. Explain the significance of Stage P.

.....

.....

.....

.....

..... [2]

- (ii) In an experiment, it was found that the concentration of the enzyme, histone deacetylase (HDAC) in infected cells at Stage **P** was higher than in infected cells not at Stage **P**.

Suggest an explanation for the presence of abnormally high levels of HDAC in infected cells at Stage **P**.

.....

.....

.....

.....

..... [2]

- (iii) With reference to Fig. 3.1, describe and explain the changes in the amount of HIV proteins and CD4+ T cells count during Stage **Q**.

.....

.....

.....

.....

..... [2]

- (iv) Suggest how viral load may be measured in the blood.

.....

.....

..... [1]

[Turn Over

- (b) HIV enters a CD4⁺ T cell. Gp120 binds to CD4 receptor on host cell surface membrane. Gp120 undergoes conformational change enabling gp120 to bind to co-receptors CCR5 on host cell membrane. Interaction between gp120 and co-receptor brings about conformational change to gp41 on viral envelop leads to fusion of HIV envelope to the host cell surface membrane, gaining entry.

Fig. 3.2 shows a CCR5 co-receptor, which is a protein.

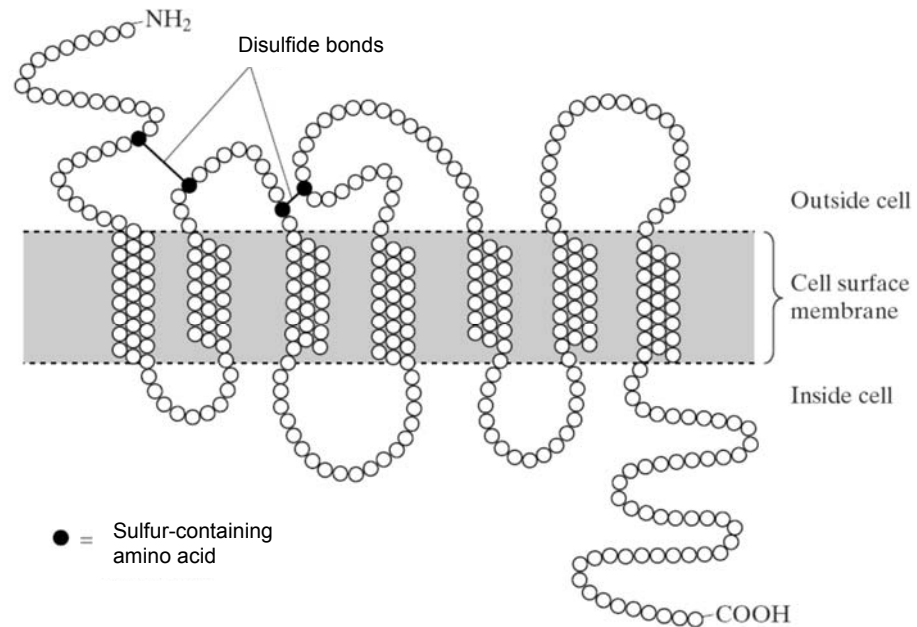


Fig. 3.2

Using information from Fig. 3.2, describe the protein structure of the CCR5 co-receptor.

.....

.....

.....

.....

..... [2]

- (c) Integrase is one of the key targets for anti-retroviral therapy. An example of an anti-retroviral drug that acts on integrase is raltegravir. Fig. 3.3 shows the normal reaction catalysed by integrase, and Fig. 3.4 shows the effect of raltegravir on integrase.

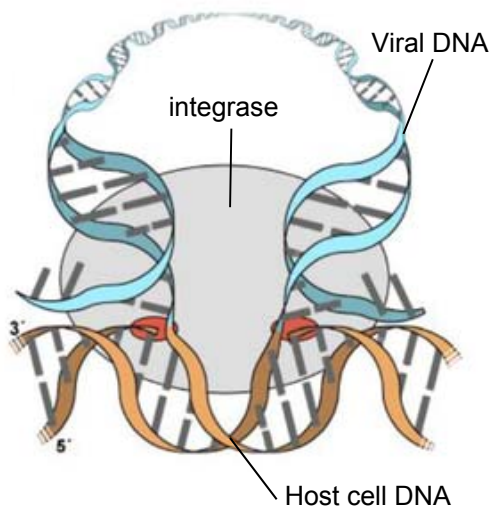


Fig. 3.3

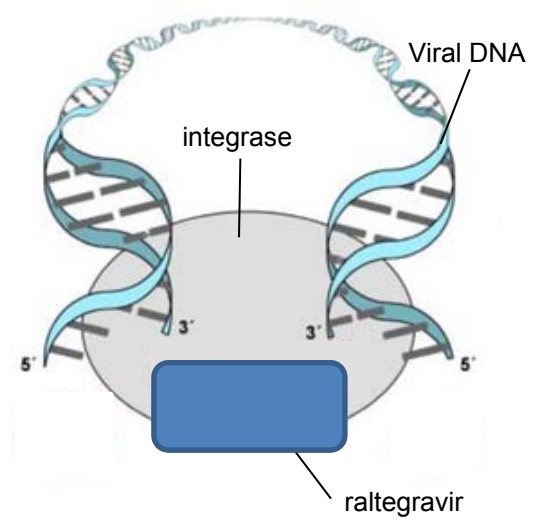


Fig. 3.4

Based on Fig. 3.3 and Fig. 3.4, describe the mode of action of raltegravir on integrase.

.....

.....

.....

.....

..... [2]

[Total: 11]

[Turn Over]

- 4 “Cancer is a disease of the genome, triggered by the accumulation of genetic errors that eventually transform a normal cell into a tumour cell. Such mutations might inactivate genes that normally oppose tumour development, or activate genes that drive cell growth or interfere with cell differentiation or death.

This leads to the 2 classes of genes which are responsible for cancer formation.”

Nature **417**, 906-907 (June 2002)

“Cancer is largely a disease of older people.”

Nature Reviews Cancer **5**, 655-662 (August 2005)

- (a) Using the above information, explain why cancer usually occurs in older individuals.

.....

.....

.....

.....

..... [2]

- (b) With specific named examples, distinguish between the 2 classes of genes stated in **line 5**.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(ii) describe how Ras may contribute towards the development of cancer.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 12]

- 5 Fig. 5.1 shows the electron micrograph of a chloroplast with structures X, Y and Z labelled.

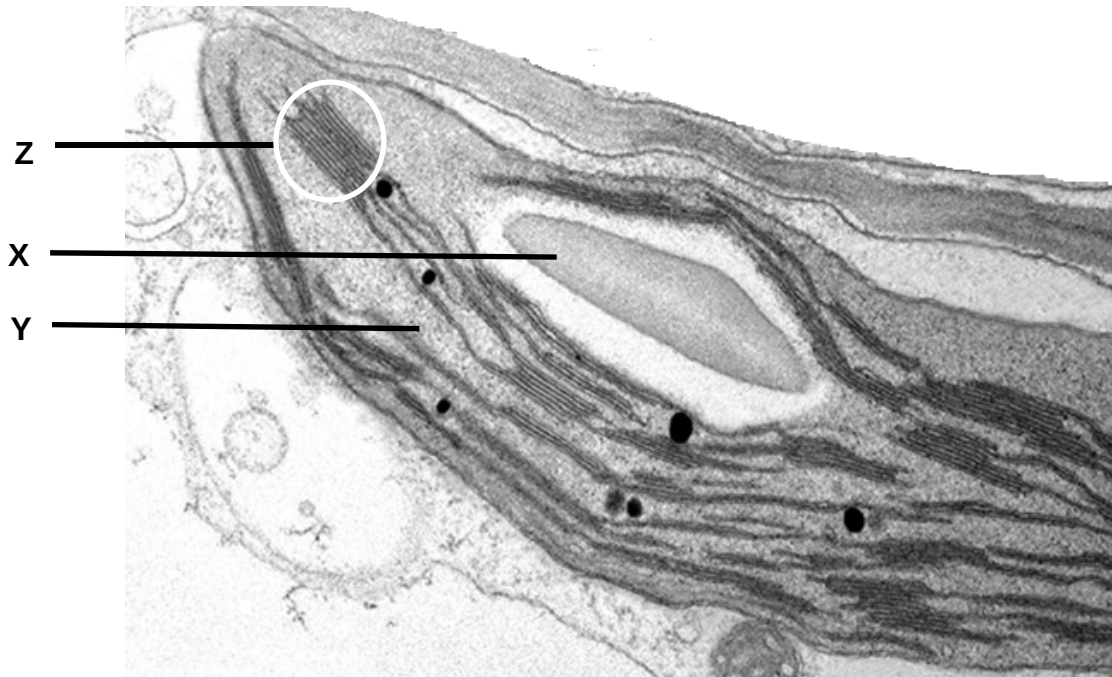


Fig. 5.1

- (a) (i) Identify structures X and Y.

X

Y [2]

- (ii) Explain how Z is adapted to carry out photophosphorylation.

.....
.....
.....
.....
..... [2]

[Turn Over]

- (b)** Rubisco is an important enzyme responsible for carbon fixation in the Calvin cycle. As Rubisco has an optimum pH of 9, it is most active during the daytime when the pH in the stroma is high.

Explain why Rubisco is most active during the daytime.

.....

.....

.....

.....

..... [2]

A process known as photorespiration also takes place in photosynthetic cells. In this process, oxygen competes with carbon dioxide for the active site of the enzyme Rubisco.

Fig. 5.2 outlines the process of photorespiration.

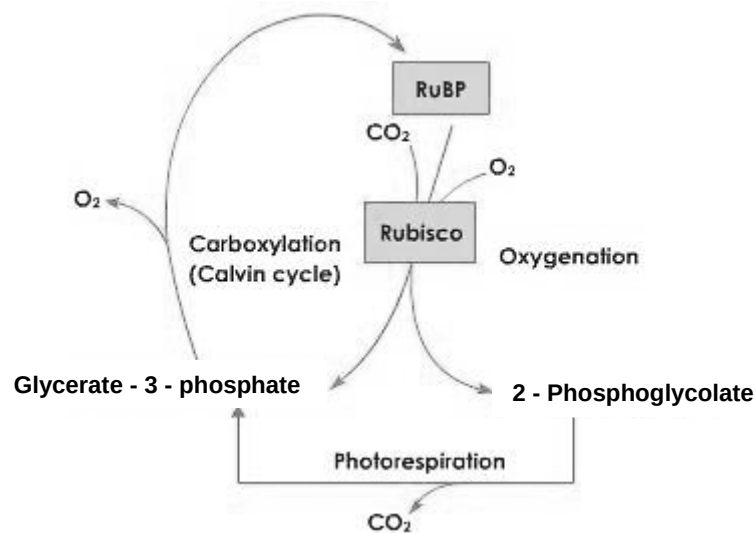


Fig. 5.2

- (c) (i)** Describe and explain the effects of an increase in oxygen concentration on photosynthesis.

.....

.....

.....

.....

.....

.....

.....
..... [3]

(ii) Suggest why the process outlined in Fig. 5.2 is known as photorespiration.

.....
.....
..... [1]

[Total: 10]

[Turn Over

- 6 An inbred variety of maize, **A**, with finely striped leaves was found to have high resistance to the fungus that causes the disease, corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

The above F_1 generation was test crossed with variety **A** and yielded the following results:

finely striped leaves and high resistance	80
finely striped leaves and low resistance	20
entirely green leaves and high resistance	22
entirely green leaves and low resistance	78

- (a) A chi-squared test was performed on the results of the cross to determine if the results of the test cross depart significantly from the expected ratio. Calculate the χ^2 value using the formula provided below.

Formula for χ^2 calculation

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad v = c - 1$$

where

Σ = 'sum of...'

v = degrees of freedom

c = number of classes

O = observed 'value'

E = expected 'value'

χ^2 value: [1]

Table 6.1

degree of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.31	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (b) Using Table 6.1, explain the conclusion drawn from the χ^2 test to determine if the observed numbers conformed to the expected.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Turn Over

(c) Using the following symbols:

G entirely green leaf

R low resistance

g

finely striped leaf

r

high resistance;

Draw a genetic diagram to show the actual results of the test cross.

[4]

[Total: 8]

- 7 (a) Fig. 7.1 is an electron micrograph of a mitochondrion in a neurone.

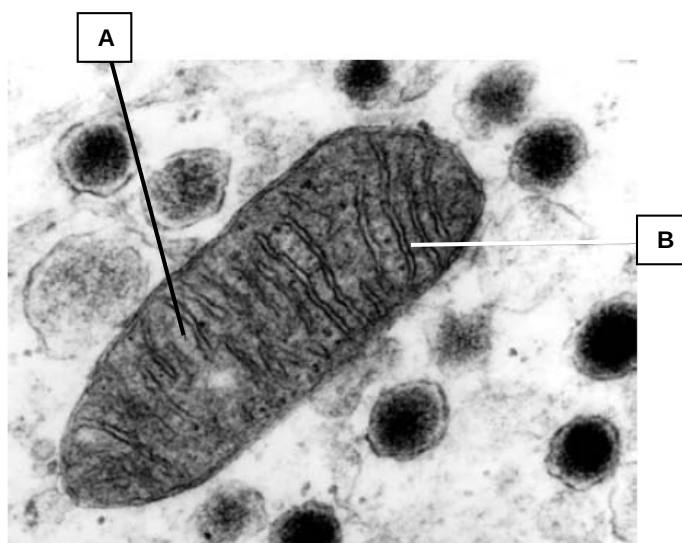


Fig. 7.1

Two stages of respiration occur in mitochondria. These are the Krebs cycle and oxidative phosphorylation.

- (i) Complete the table below by naming the structures labelled **A** and **B** and stating which of the stages of respiration occur in each.

	Name of structure	Stage of respiration
A		
B		

[2]

[Turn Over

(ii) Explain the need for mitochondria along the axon in terms of nerve impulse conduction.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Serotonin is a neurotransmitter which is produced by certain neurones in the brain. One of its effects is to increase the activity of sensory neurones in the brain. It also usually improves a person's mood and keeps the person awake. Fig. 7.2 shows a synapse at which serotonin is the neurotransmitter.

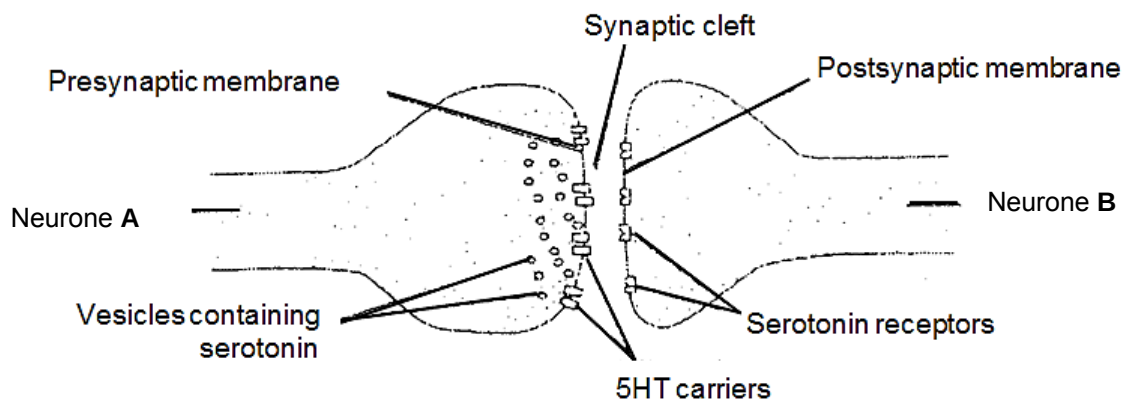


Fig. 7.2

(i) Describe how the neurotransmitter is released into the synaptic cleft.

.....

.....

.....

.....

.....

..... [2]

- (ii) Explain how the release of the neurotransmitter, serotonin, results in the transmission of an impulse from neurone **A** to neurone **B**.

.....

.....

.....

.....

..... [2]

- (iii) Suggest why nerve impulse can only travel in one direction across a synapse.

.....

.....

..... [1]

[Total: 10]

[Turn Over]

- 8 There are over 40 Galapagos Islands including the small and isolated island named Daphne Major.

Studies were made every year from 1970 to 1989 on the beak size of the island's population of ground finch, *Geospiza fortis*, by measuring the beak length of every bird (Fig. 8.1). Larger finches with larger beaks are better at opening large seeds.



Fig. 8.1

From 1976 to 1978 there was a drought and only 15% of the ground finches survived and these did not breed during drought years. The most conspicuous feature of the survivors of the drought years was their large beak size.

During normal years, many drought-intolerant grasses and herbs produce an abundance of small seeds. A few other drought-tolerant plants produce a much smaller number of large seeds which are not normally eaten.

- (a)** Describe how environmental factors act as forces of natural selection, during drought years, on the beak size of finches.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) Suggest the role of the islands in the evolution of thirteen species of Darwin finches now found on the Galapagos Islands.

.....

.....

.....

.....

..... [2]

Molecular analysis was carried out on the mitochondrial DNA (mtDNA) sequences of the Galapagos Islands finches and the Cocos finch found on the island of Cocos, 830 km to north-east of the Galapagos Islands. Using mtDNA analysis data, a map showing the phylogeny of these finches was constructed as shown below in Fig. 8.2.

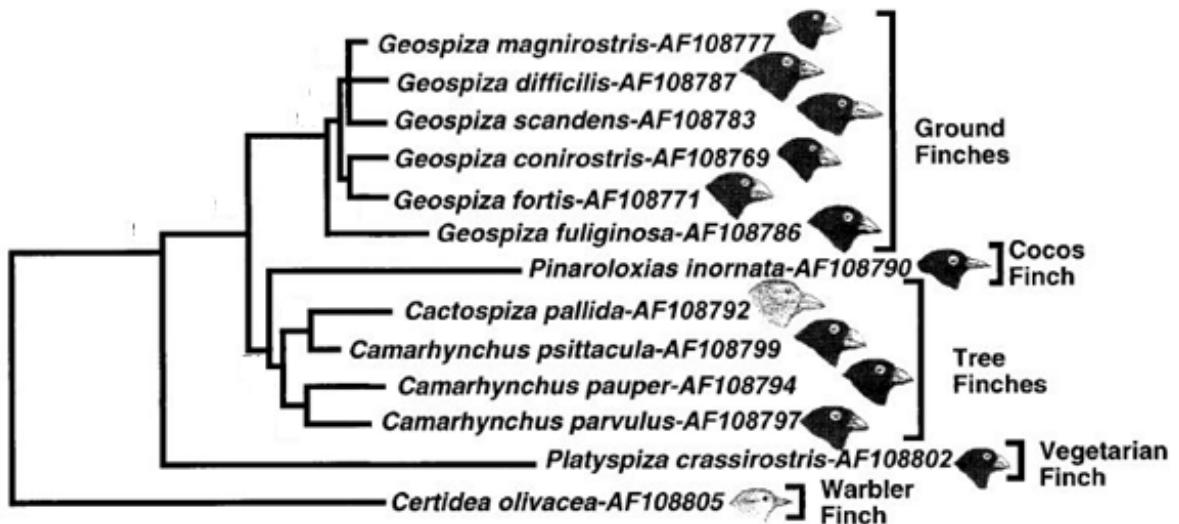


Fig. 8.2

- (c) Explain how DNA sequences can be used to determine evolutionary relatedness between species.

.....

.....

.....

.....

..... [2]

[Turn Over

(d) Describe the advantages of using nucleotide data such as mtDNA in classifying organisms.

.....

.....

.....

.....

..... [2]

Differences in the *cytochrome b* DNA sequence of several finches' species from Galapagos Islands and island of Cocos were measured and plotted against time since divergence from the primitive ancestor (MYA) as seen in Fig. 8.3.

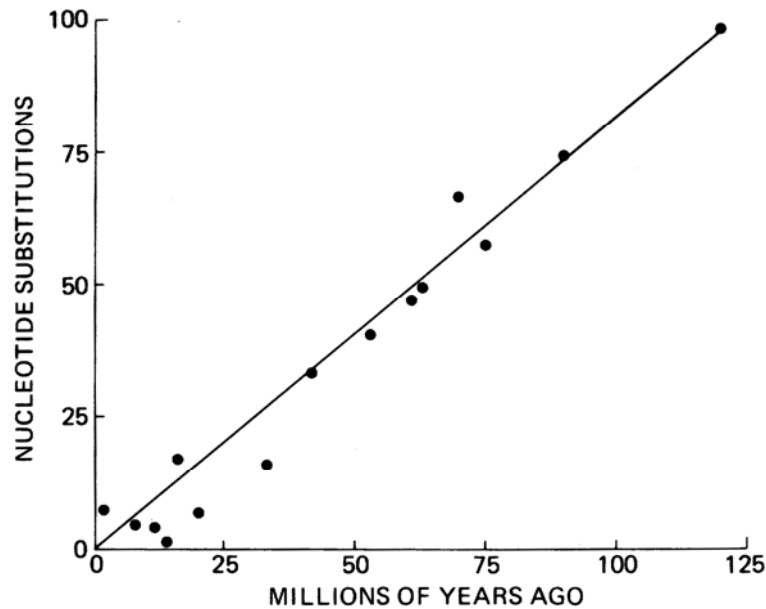


Fig 8.3

(e) Describe how these differences support the neutral theory of molecular evolution.

.....

.....

.....

.....

..... [2]

[Total: 12]

Section B

Answer **one** question.

Write your answers to this question on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9 (a)** Compare and contrast between glucagon and glycogen. [4]
- (b)** Explain what is meant by primary, secondary, tertiary, and quaternary structure of a named protein. [8]
- (c)** Describe the main properties of an enzyme and discuss their mode of action. [8]

[Total: 20]

- 10 (a)** With reference to the islets of Langerhans, describe what is meant by an endocrine gland. [7]
- (b)** Explain how the blood glucose concentration is regulated by insulin and glucagon. [8]
- (c)** Describe how the endocrine system differs from the nervous system. [5]

[Total: 20]

[Turn Over]

BLANK PAGE